

KEERTHI KUMAR KARANI

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PROFESSIONAL SUMMARY:

- **Senior ETL Developer** with 8 years of experience architecting, developing, and optimizing enterprise data pipelines and cloud-based data warehouse solutions across insurance, healthcare, telecom, and HRMS domains, processing 100M+ records/day at production scale.
- Deep hands-on expertise in **Informatica PowerCenter**, designing complex mappings, mapplets, workflows, and session configurations for large-scale data integration across SQL Server, Oracle, Teradata, flat files, and cloud endpoints across regulated healthcare and telecom environments.
- Engineered **ETL solutions** using Python, SQL, Shell scripting, Informatica PowerCenter, Ab Initio, SSIS, DataStage, Azure Data Factory, Azure Databricks, and Snowflake — modernizing legacy infrastructure to resolve data fragmentation, reporting delays, and compliance gaps across regulated enterprises.
- Drove **Lakehouse adoption** by implementing Bronze/Silver/Gold architecture using Delta Lake and PySpark, consolidating fragmented source systems into unified analytical layers — eliminating ~60% of enterprise data silos that had blocked cross-brand fraud detection and workforce analytics.
- Delivered **enterprise-scale Teradata-to-Snowflake migrations** using AWS Glue, PySpark, and automated reconciliation scripts — reconciling TB-scale historical datasets with 99%+ accuracy and cutting migration runtimes by 40%.
- Accelerated claims adjudication and payroll processing cycles by implementing **CDC, incremental loading, SCD Type 1/Type 2 strategies, and Spark/SQL performance tuning** — eliminating full table reloads and cutting pipeline processing time by up to 45% across SLA-critical workflows.
- Implemented **data quality validation frameworks** using PySpark and reconciliation checks those intercepted anomalies before downstream consumption — reducing reporting errors by ~30% and restoring stakeholder confidence in claims and payroll data.
- Achieved **audit-ready status** across HIPAA, PCI-DSS, and SOC 2 regulated environments by enforcing RBAC, field-level encryption via Azure Key Vault, dynamic data masking, and PHI protection — sustaining zero unauthorized access incidents across 3+ audit cycles.
- **Orchestrated fault-tolerant ETL workflows** using Airflow, Databricks Jobs, and ADF with automated retry logic, SLA alerting, and dependency chaining — sustaining 99.5%+ pipeline reliability while accelerating deployment cycles by ~35% through CI/CD via Azure DevOps and Jenkins.
- Delivered **analytics-ready datasets** powering Power BI and Tableau dashboards for fraud detection, claims tracking, workforce attrition, and customer churn — enabling leadership to identify fraud patterns 3x faster and cut decision-making turnaround by 40%.
- **Led and mentored teams** of 3–5 ETL developers across onshore/offshore delivery models, conducted technical design reviews, established ETL coding standards, and partnered with actuarial, HR operations, finance, and marketing stakeholders to translate complex business requirements into scalable, governed data solutions.
- Executed **Ab Initio to PySpark and Databricks conversion** initiatives, translating Ab Initio graph-based ETL logic into modular PySpark jobs on Databricks — reducing legacy licensing costs by ~35% while improving pipeline scalability and enabling cloud-native execution across regulated enterprise environments.

TECHNICAL SKILLS:

Languages	Python (PySpark, Pandas, NumPy), SQL, Unix Shell Scripting
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ETL & Data Integration	Informatica PowerCenter, Ab Initio (GDE, Co>Operating System), Azure Data Factory, Databricks (PySpark, Delta Live Tables), PySpark, SSIS, Talend, DataStage, Delta Lake, Apache Spark, Apache Kafka
Data Warehousing & Modeling	Snowflake, Teradata, SQL Server, Oracle, PostgreSQL, MySQL; Dimensional Modeling, SCD Type 1 & Type 2, Data Migration, Erwin, Lucid Chart
Cloud Platforms	Azure (ADF, Databricks, ADLS, Synapse Analytics, Cosmos DB), AWS (S3, EC2, Redshift, Glue)
Workflow Orchestration & DevOps	Airflow, Databricks Jobs, Git, Jenkins, Azure DevOps, CI/CD pipelines
Databases	Snowflake, Teradata, Oracle, Microsoft SQL Server, MySQL, PostgreSQL, Db2, T-SQL, Query Optimization, Stored Procedures, Functions
Data_Quality Governance	& Data Validation, Data Reconciliation, Metadata Management, Lineage Tracking, Data Standards, Audit-Ready Frameworks
Security & Compliance	RBAC, Data Encryption (at-rest & in-transit), Azure Key Vault, Dynamic Data Masking, HIPAA, PCI-DSS, SOC 2, PHI Protection
Visualization Analytics:	& Power BI, Tableau, Excel (Advanced), KPI Reporting, Dashboard Development
Functional Skills	ETL Development, Predictive Analytics, Data Analysis, Data Visualization, Time Series Forecasting, Data Modeling, Data Pipeline Design, Data Integration, Data Management
Statistical & Analytical Methods	Regression Analysis, Time Series Analysis, Hypothesis Testing, A/B Testing, Exploratory Data Analysis
Operating Systems	Windows, Linux/Unix-like systems

CERTIFICATIONS:

Azure Certified, Fundamentals

Azure Certified, Data Fundamentals

Azure Certified, Fabric Analytics Associate

Azure Certified, Fabric Data Engineer Associate

Google Certified, Engineer Data in Cloud

PROFESSIONAL EXPERIENCE:

Client: Assurant FL

Role: Sr ETL Developer

Responsibilities:

Jul'25 – Present

- Re-engineered legacy Ab Initio and DataStage workflows into **Informatica PowerCenter pipelines**, consolidating TB-scale warranty claims and policy data across 3+ siloed business units — eliminating ~60% of data silos and enabling unified cross-brand fraud detection.
- Designed **Informatica PowerCenter mappings** using **Joiner, Lookup, Router, Aggregator, and Expression transformations** to implement CDC and incremental loading — cutting pipeline processing time by ~45% and compressing claims adjudication from 72 to under 36 hours.
- Built reusable **Informatica mapplets enforcing SCD Type 2** on policy and warranty dimension tables in Snowflake — enabling actuarial point-in-time analysis previously impossible under the overwrite-only SCD Type 1 approach.
- Tuned **Informatica mapping performance** by converting unconnected lookups to connected, pushing logic to source qualifiers, and optimizing cache settings — reducing session run time by ~25% across critical claims pipelines.
- Configured **Informatica session partitioning (hash and key-range)** across high-volume claims workflows — improving pipeline throughput by 30% and sustaining processing of 50M+ daily records without bottlenecks.
- Deployed **Informatica-based data quality validation framework** with reconciliation checks targeting downstream reporting errors — cutting data error rate by ~30% and improving claims accuracy to 99%+ across compliance datasets.
- Automated **end-to-end Informatica workflow scheduling** with dependency chaining, retry logic, and ADF orchestration — sustaining pipeline uptime at 99.5%+ and reducing MTTR by 40%.
- Conducted **Snowflake query profiling** on Informatica-loaded datasets to resolve inefficient joins and suboptimal clustering keys — reducing compute cost per query by 25% and improving warehouse credit utilization by 30%.
- Enforced **RBAC, Azure Key Vault encryption, and PII/PHI masking** across all Informatica pipelines — achieving zero unauthorized access incidents across 3 PCI-DSS audit cycles.
- Delivered **Power BI dashboards on Informatica-curated Snowflake** datasets tracking claims volume, denial rates, and fraud hit rate — enabling leadership to identify fraud patterns 3x faster.
- **Led 4 ETL developers** across onshore/offshore delivery, establishing reusable Informatica mapping templates and coding standards — reducing development rework by ~20% and accelerating sprint velocity.
- Assessed and documented **legacy Ab Initio GDE graphs** ingesting warranty claims and policy data — cataloging 40+ Ab Initio components, mapping transformation logic, and building a conversion blueprint that served as the foundation for phased migration to PySpark and Databricks.
- Converted 15+ **Ab Initio batch ETL jobs to PySpark on Azure Databricks** using Delta Live Tables and Delta Lake — achieving functional parity with a 30% reduction in end-to-end processing time and cutting Ab Initio licensing overhead by ~35% on migrated workloads.

Environment: Informatica PowerCenter, Ab Initio, Snowflake, Azure Data Factory, PySpark, Databricks, ADLS, Python, Azure Key Vault, Azure DevOps, Delta Lake, PCI-DSS, SCD Type 1 & Type 2

Client: Community Dreams Foundation

Jul '23 – Jul '25

Role: ETL Developer

Responsibilities:

- Architected **Informatica PowerCenter** pipelines ingesting HRMS, payroll, and employee data from 8+ disconnected source systems (SQL Server, Oracle, flat files) into Snowflake — resolving chronic payroll-to-headcount mismatches and bringing reconciliation accuracy to ~98.5%.

- Designed **Informatica source qualifier overrides** and pre/post-session SQL commands to handle payroll period-end boundary conditions — eliminating duplicate pay records that were causing ~3% payroll variance downstream.
- Built **Informatica mapplets** with Router and Expression transformations to split employee records by employment type (full-time, part-time, contract) and apply type-specific business rules — standardizing data across 8+ HR source systems with inconsistent schemas.
- Implemented **SCD Type 2 logic** in Informatica for compensation dimensions (salary bands, job grades, designation history) using effective date flagging — enabling HR to accurately track historical compensation changes for regulatory audits.
- Used **Informatica's dynamic lookup** cache on employee master data to handle real-time attribute updates during payroll runs — preventing stale dimension lookups that had caused incorrect payroll period allocations.
- Developed **parameterized Informatica workflows** with variable substitution to support multi-tenant HR data loads across different business units — reducing new department onboarding effort from weeks to days.
- Performed **Informatica repository-level mapping** analysis to identify and consolidate 40+ redundant mappings built by prior developers — reducing maintenance overhead and improving deployment consistency across environments.
- Configured **Informatica grid processing and session recovery settings** on high-volume monthly payroll loads — ensuring zero data loss on session failures and cutting full-reload recovery time by ~35%.
- Enforced **column-level data masking** on PII fields (SSN, salary, bank details) within Informatica mappings before loading into Snowflake analytical layers — satisfying SOC 2 auditor requirements and maintaining compliance score at 100%.
- Delivered **Power BI dashboards** on Informatica-curated Snowflake datasets surfacing FTE variance, cost-per-hire, attrition rate, and retention risk — enabling HR leadership to cut time-to-fill by ~20% and improve attrition prediction accuracy by ~25%.
- **Mentored 2 junior developers** on Informatica session configuration, mapping debugging, and incremental load design — cutting their onboarding ramp-up from 8 weeks to under 5 weeks.
- Piloted **migration of high-volume Informatica payroll workflows to PySpark on Databricks** using Delta Lake and ADF orchestration — validating functional parity with 99%+ reconciliation accuracy and establishing a reusable conversion pattern for broader legacy ETL modernization across the organization.
- Translated complex Informatica mapplet logic into modular **PySpark transformation functions on Databricks** — implementing SCD Type 2 compensation history and multi-tenant HR partitioning natively in Spark, reducing batch window by 40% compared to the Informatica baseline.

Environment: Informatica PowerCenter, Azure Data Factory, ADLS, Databricks, PySpark, Delta Lake, Snowflake, SQL Server, Oracle, Python, Power BI, Tableau, SOC 2, SCD Type 1 & Type 2

Client: AT&T, DirecTV- (AMDOCS)

Nov '21 – Jun '23

Role: Business Intelligence Developer/ETL

Responsibilities:

- Engineered and optimized **PySpark ETL pipelines** on AWS EMR processing 1M+ daily telecom and marketing transactions across AT&T and DirecTV brands to resolve ingestion bottlenecks — improving data ingestion throughput by 40% and reducing end-to-end pipeline latency from hours to minutes.
- Spearheaded Teradata-to-Snowflake migration using **AWS Glue and PySpark** to move off a legacy platform — reconciling 3+ years of historical data with data reconciliation accuracy of 99%+ and cutting migration runtime by 40%.

- Designed **dimensional models** using AWS Glue-processed datasets with partitioning and indexing to address slow dashboard load times — improving Power BI query response time by 35% and bringing report load latency from minutes to under 5 seconds.
- Implemented **SCD Type 2** on customer and subscriber dimension tables to preserve historical subscription states, plan changes, and account transitions — enabling marketing analytics teams to perform accurate cohort analysis and churn trend reporting across multi-year timeframes.
- Configured **CDC and incremental load strategies** using AWS DMS and AWS Glue to replace nightly batch-only synchronization — reducing data latency by ~50% and improving data currency SLA from T+24 hours to near real-time for customer account synchronization.
- Built scalable **real-time ingestion workflows** using AWS Kinesis Streams to process telecom events and customer transactions — enabling sub-second data availability and improving streaming throughput to handle peak transaction volumes without backpressure.
- Deployed robust **data validation and monitoring frameworks** that proactively surfaced ETL failures before they reached downstream consumers — reducing pipeline failure rate by 20% and improving **pipeline reliability SLA to 95%+.**
- Implemented **data lake architecture** on AWS S3 using a layered pattern with AWS Glue Data Catalog — improving data discoverability score by 30% and reducing average time-to-data for analysts from hours to minutes.
- **Collaborated with marketing, product, and data science teams** across AT&T and DirecTV brands to define KPI definitions, segmentation logic, and campaign attribution models — ensuring ETL pipeline output aligned with cross-brand reporting requirements and business decision timelines.
- Resolved recurring EMR pipeline failures through **root cause analysis** identifying data skew and inefficient joins — reclaiming ~6 hours/week of engineering bandwidth and reducing incident escalation volume by 20%.

Environment: AWS EMR, AWS Glue, AWS S3, AWS Kinesis, AWS DMS, AWS Redshift, PySpark, Snowflake, Power BI, Teradata, GitHub, Jenkins, Erwin, Lucid Chart, SCD Type 2

Client: United Health Group, T-Mobile – (Infosys)

May '18 – Oct '21

Role: Senior Systems Engineer (ETL Developer)

Responsibilities:

- Developed and optimized **SSIS and Informatica ETL pipelines** ingesting 1M+ daily healthcare and telecom transactions from SQL Server, Oracle, Teradata, and flat files — stabilizing the ingestion layer and improving SLA adherence from ~80% to 95%+ across all pipeline schedules.
- Redesigned Teradata **dimensional models** with partitioning, indexing, and HIPAA-compliant governance controls to resolve underperforming analytical queries — delivering 35% faster query retrieval time and reducing average report generation time by 40%.
- Implemented **SCD Type 1** for frequently updated patient demographics and provider attributes, and **SCD Type 2** for audit-critical dimensions including diagnosis history, insurance plan transitions, and billing code changes — enabling compliant historical reporting required for CMS submissions and HIPAA audit trails.
- Performed Teradata **query performance tuning** including index optimization, statistics collection, and join redistribution to resolve long-running clinical and billing queries — reducing average query execution time by 30% and improving concurrent query throughput by 25%.
- Designed **HIPAA-compliant** EHR/EMR dashboards using Power BI, Tableau, and Excel tracking patient throughput, readmission rates, length-of-stay, and clinical utilization KPIs — increasing reporting efficiency by 35% and eliminating ~8 hours/week of manual report production.

- **Built real-time healthcare monitoring dashboards** using Azure Synapse Analytics and Power BI to replace end-of-day batch reporting — reducing clinical decision-making latency by 40% and improving KPI refresh frequency from daily to near real-time.
- Developed T-Mobile **customer churn dashboards** and predictive feature datasets using Power BI and Python tracking churn rate, customer lifetime value (CLV), and retention campaign ROI — enabling targeted campaigns that reduced monthly churn rate by 15%.
- Led **Teradata-to-Snowflake migration** of TB-scale historical data resolving data quality issues the prior team had deprioritized — achieving data reconciliation accuracy of 99.9%+ and completing migration ahead of schedule.
- Developed **HL7, ORU, and ADT message ingestion workflows** to enable seamless integration between hospital information systems and downstream clinical analytics platforms — achieving message processing success rate of 99%+ and reducing integration error rate by 25%.
- Enforced HIPAA-compliant **RBAC and field-level encryption** across ADF and Azure Data Lake pipelines to close compliance gaps flagged in prior audit cycles — reducing operational compliance error rate by 20% and achieving audit readiness score of 100%.
- Automated **ETL monitoring and job orchestration** using Python scripts to replace manual overnight oversight — improving SLA adherence by 20%, reducing mean time to detection (MTTD) of failures by 30%, and cutting manual oversight by 25%.
- **Led a team** of 3 junior ETL developers conducting code reviews, knowledge transfer sessions, and design walkthroughs on SCD implementation, Informatica mapping optimization, and Teradata performance tuning — reducing onboarding ramp-up time by ~30% and improving team delivery consistency.
- **Collaborated** with clinical operations, billing, and product teams to define data requirements, KPI specifications, and compliance reporting standards — ensuring ETL deliverables aligned with CMS submission timelines and business analytics priorities.

Environment: Informatica PowerCenter, SSIS, Azure Data Factory, Azure Data Lake, Python, SQL, Teradata, SQL Server, Power BI, Unix, Tableau, Jenkins, HIPAA, RBAC, SCD Type 1 & Type 2